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MBA Tech
DS
Sem - 3

Assignment - 2

Q.1

$$^{52}C_4 = \frac{52!}{4!\,48!} = 270725$$

(i) ~~There are a King,~~ Queen, Jack, & Ace. There are four kings, 4 queens, 4 Jacks & 4 aces.

Favourable ways : $4 \cdot 4 \cdot 4 \cdot 4 = 4^4 = 256$

$\therefore P = \frac{256}{^{52}C_4} = \frac{256}{270725}$ or ≈ 0.000946

Ans

(ii) Two are Kings & two are aces

Choose 2 kings from 4 & 2 Aces from 4.

Favourable ways $= {}^4C_2 \times {}^4C_2 = 6 \cdot 6 = 36$

$\therefore P = \frac{36}{^{52}C_4} \Rightarrow P = \frac{36}{270725} \approx 0.000133$

Ans

(iii) All four are diamonds

There are 13 diamonds.

Favourable ways

$^{13}C_4$

$P = \frac{^{13}C_4}{^{52}C_4} = \frac{715}{270725} \approx 0.00264$

Ans

(iv) Two are red & two are black

There are 26 red & 26 black cards.

Choose 2 red & 2 black

Favourable ways = ${}^{26}C_2 \times {}^{26}C_2$
$= 325 \times 325 = 105625$

$\therefore P = \frac{105625}{270725} \approx 0.3903$ Ans

(v) There ~~are~~ is one card of each suite'
There are 4 suits, each containing 13 cards

choose one card from each suit:
$13 \cdot 13 \cdot 13 \cdot 13 = 13^4 = 28561$

Thus, $P = \frac{13^4}{{}^{52}C_4} = \frac{28561}{270725} \approx 0.1055$ (Ans)

(vi) Two cards are clubs & two are diamonds
Favourable ways = ${}^{13}C_2 \times {}^{13}C_2 = 78 \times 78 = 6084$ (Ans)

$\therefore P = \frac{6084}{270725} \approx 0.02247$ (Ans)

Q.2 Total tickets: $n(S) = 20$
The multiple of 2 or 5:
Multiples of 2:
2, 4, 6, 8, 10, 12, 14, 16, 18, 20
Number = 10

Multiples of 5:
~~50~~ 5, 10, 15, 20.
Number = 4

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numbers common in both : 10, 20

Number = ~~10~~ 2

Using Inclusion - Exclusion principle:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 10 + 4 - 2 = 12$

$\therefore$ $P(2 \text{ or } 5) = \frac{12}{20}$

$P = 0.6$ (Ans)

(ii) Multiple of 3 or 5

Multiples of 3:

3, 6, 9, 12, 15, 18

number = 6

Multiples of 5 : 5, 10, 15, 20

number = 4

Common : 15 $\Rightarrow$ number = 1

$\therefore$ $n(3 \text{ or } 5) = 6 + 4 - 1 = 9$

$\therefore$ $P = \frac{9}{20} = 0.45$ (Ans).

Q.3

$P(x|o) = \dfrac{P(x \cap o)}{P(o)}$

from the table : $P(x \cap o) = 0.125$

& $P(o) = 0.125 + 0.0625 = 0.1875$

$\therefore P(x|o) = \dfrac{0.125}{0.1875}$; $P(x|o) = 0.6667$

(Ans)

(ii) Part was produced by x. Find probability that it has no defects

$P(o|x) = ?$

$P(x) = 0.125 + 0.0625 + 0.1875 + 0.125$

$P(x) = 0.5$

Therefore,

$$P(o|x) = \frac{P(x \cap o)}{P(x)} = \frac{0.125}{0.5}$$

$P(o|x) = 0.25$ (Ans)

(iii) Part has two or more defects. Find probability it was manufactured by x.

We need : $P(x|2+)$

for x: $P(x \cap 2+) = P(x \cap 2) + P(x \cap 3)$

$= 0.1875 + 0.125 = 0.\frac{8}{25}$

$\therefore P(2+) = 0.3125 + 0.375 = 0.6875$

$P(x|2+) = \dfrac{0.3125}{0.6875} = 0.4545$

Ans

(iv) $P(Y|1+)$

for $P(Y \cap 1+) = 0.0625 + 0.125 + 0.250$
$= 0.4375$

for 'X':

$P(X \cap 1+) = 0.0625 + 0.1875 + 0.125$
$= 0.375$

$P(1+) = 0.4375 + 0.375 = 0.8125$

therefore:

$P(Y|1+) = \frac{0.4375}{0.8125}$

$P(Y|1+) = 0.5385$ (Ans)

Q.4 Government:

$5 + 8 + C = 13$
$C = 0$

Academics:

$150 - 22 - 13 = 65$
$\therefore 25 + E + 20 = 65$
$E = 20$

Pvt. Sector: $S + 7 + C = 22$
Total stable is 40
$25 + S + 5 = 40$
$S = 10$

$\therefore 10 + 7 + C = 22$
$C = 5$

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Total expanding : 20 + 7 + 8 = 35
Total contracting : 20 + 5 + 0 = 25

So the completed table is :

Economist	Stable	Expanding	Contracting	Total
Acad (A)	25	20	20	65
Pvt. Sector (R)	10	7	5	22
Government (G)	5	8	0	13
Total	40	35	25	100

(i) $P(A) = \frac{\text{Acad}}{\text{Total}} = \frac{65}{100} = 0.65$ Ans

(ii) $P(E) = \frac{\text{Pvt. Sector}}{\text{Total}} = \frac{35}{100} = 0.35$ Ans

(iii) $P(A \cap E)$

$P(A \cap E) = \frac{20}{100} = 0.20$ Ans

(iv) $P(G \cap O) = \frac{0}{100} = 0$ Ans

(v) Find $P(E|G) = \frac{P(E \cap G)}{P(G)} = \frac{8}{13}$

$= 0.6154$ (Ans)

(vi) $P(G|E) = \frac{P(G \cap E)}{P(E)} = \frac{8}{35} = 0.2286$ Ans

Q.5 Households with monthly income of ₹ 8000 or less | below ₹ 8000

	Tel.	No Tel.	Tel.	No Tel.
Own TV	27	20	18	10
No TV	18	10	12	10

Total household : 27 + 10 + 18 + 18 + 10 + 12 + 10 = 125

(i) Probability of obtaining of a TV owner is drawing at random

Total no. of TV owners = 27 + 20 + 18 + 10 = 75

$P(A) = 75/125$

$P(A) = 0.6$ $P(B) = 1/5$

~~$P(A \text{ or } B) =$~~ $P(TV) = \cancel{0.6}$ $\frac{75}{125} = 0.6$ Ans

(ii) $P(TV \mid I > 8000, \text{telephone}) = 0.6$ Ans

(iii) $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$P(TV \mid \text{telephone}) = 0.6$ Ans

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(iv) $P(A|B) = P(A)$

$P(TV \mid telephone) = 0.6$

& $P(TV) = 0.6$

$\therefore P(TV \mid telephone) = P(TV)$

$\rightarrow$ <u>Independent</u>

Q.6 Total = 9 (2 left handed + 7 right handed)

(i) (a) probability that both gloves are right handed without replacement

First glove is right handed =

$P(R_1) = \frac{7}{9}$

$P(R_2 | R_1) = \frac{6}{8}$

$\therefore P(R_1 \cap R_2) = \frac{7}{9} \times \frac{6}{8}$

$\frac{42}{72} = \frac{7}{12} \therefore P = \frac{7}{12} = 0.5833$

(i) (b) $P(LR) = \frac{2}{9} \times \frac{7}{8}$

$= \frac{7}{36}$

$P(RL) = \frac{7}{36}$ $\therefore P(LR \text{ or } RL) = \frac{7}{36} + \frac{7}{36}$

$= \frac{7}{18} = 0.3$

(ii) There are only 2 left handed gloves, it is impossible to select 3 left-handed gloves

Therefore : $P = 0$

c(iii) $P(R) = \frac{7}{9}$ for both selections

$P(RR) = \frac{7}{9} \cdot \frac{7}{9} = \frac{49}{81}$ Ans

Q.7 Bag 1 : 5 white balls

3 black

Total = 8

Bag 2 : 4 white balls

5 black

Total = 9

$P(B_1) = P(B_2) = \frac{1}{2}$

$P(WB) = \frac{5}{8} \times \frac{3}{7}$

or $P(BW) = \frac{3}{8} \times \frac{5}{7}$

$\therefore \frac{15}{56} + \frac{15}{56} \Rightarrow \frac{30}{56} = \frac{15}{28}$

P(one each | B_1) = 0.5357

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Probability of Bag 2

Bag 2 has 4 white & 5 black

$P(\text{one each} \mid B_2) = \frac{4}{9} \times \frac{5}{8} + \frac{5}{9} \times \frac{4}{8}$

$= \frac{20}{72} + \frac{20}{72} = \frac{40}{72} = 5/9$

$P(\text{one each} \mid B_2) = 0.5556$ Ans

Overall Probability:

$P(\text{one each}) = \frac{1}{2}\left\{\frac{15}{28}\right\} + \frac{1}{2}\left\{\frac{5}{9}\right\}$

$= \frac{15}{56} + \frac{5}{18}$

Taking LCM: $\frac{135 + 140}{504}$

$= \frac{275}{504} \approx \underline{0.545}$ (Ans)

$\frac{R}{13/8}$